

denote the corresponding approximation process, cf. Assumption 1.1(r) part (2). Then, by the reverse triangle inequality,

$$\begin{aligned}
\mathbb{E}[|X_0 - X_0^{(m)}|^2] &= \mathbb{E}[|W_r^2(\text{PD}(\mathcal{X}_0), \mu_{\mathcal{X}}) - W_r^2(\text{PD}(\mathcal{X}_0^{(m)}), \mu_{\mathcal{X}})|^2] \\
&\leq \mathbb{E}\left[\{2W_r(\text{PD}(\mathcal{X}_0), \mu_{\mathcal{X}}) + W_r(\text{PD}(\mathcal{X}_0^{(m)}), \text{PD}(\mathcal{X}_0))\}^2 \right. \\
&\quad \left. \cdot |W_r(\text{PD}(\mathcal{X}_0), \mu_{\mathcal{X}}) - W_r(\text{PD}(\mathcal{X}_0^{(m)}), \mu_{\mathcal{X}})|^2\right] \\
&\leq \mathbb{E}\left[4W_r^2(\text{PD}(\mathcal{X}_0), \mu_{\mathcal{X}}) + 2W_r^2(\text{PD}(\mathcal{X}_0^{(m)}), \text{PD}(\mathcal{X}_0))\} \cdot W_r^2(\text{PD}(\mathcal{X}_0), \text{PD}(\mathcal{X}_0^{(m)}))\right] \\
&= 4\mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \mu_{\mathcal{X}})]^{1/2} \cdot \mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \text{PD}(\mathcal{X}_0^{(m)}))]^{1/2} \\
&\quad + 2\mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \text{PD}(\mathcal{X}_0^{(m)}))].
\end{aligned}$$

Consequently, (since $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ for $a, b \geq 0$),

$$\begin{aligned}
\sum_{m \geq 1} \mathbb{E}[|X_0 - X_0^{(m)}|^2]^{1/2} &\leq 2\mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \mu_{\mathcal{X}})]^{1/4} \sum_{m \geq 1} \mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \text{PD}(\mathcal{X}_0^{(m)}))]^{1/4} \\
&\quad + \sqrt{2} \sum_{m \geq 1} \mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \text{PD}(\mathcal{X}_0^{(m)}))]^{1/2}.
\end{aligned}$$

By assumption $\mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \mu_{\mathcal{X}})]$ is finite and thus it remains to establish

$$\sum_{m \geq 1} \mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \text{PD}(\mathcal{X}_0^{(m)}))]^{1/4} < \infty.$$

To that end we use Lemma 3.10, which implies the following fundamental inequality between the Wasserstein distance W_r , resp., W_B of two diagrams $\text{PD}(U), \text{PD}(V)$ and the Hausdorff distance d_H of the underlying point clouds U, V : If $r < \infty$, then

$$\begin{aligned}
W_r(\text{PD}(U), \text{PD}(V)) &\leq W_B(\text{PD}(U), \text{PD}(V)) \cdot \max\{\#\text{PD}(U), \#\text{PD}(V)\}^{1/r} \\
&\leq 2d_H(U, V) \cdot \max\{\#\text{PD}(U), \#\text{PD}(V)\}^{1/r}.
\end{aligned}$$

Thus, we have for any Hölder conjugates $p, q \geq 1$

$$\begin{aligned}
&\mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \text{PD}(\mathcal{X}_0^{(m)}))]^{1/4} \\
&\leq 2 \mathbb{E}[d_H^4(\mathcal{X}_0, \mathcal{X}_0^{(m)}) \cdot \max\{\#\text{PD}(\mathcal{X}_0), \#\text{PD}(\mathcal{X}_0^{(m)})\}^{4/r}]^{1/4} \\
&\leq 2 \mathbb{E}[d_H^{4p}(\mathcal{X}_0, \mathcal{X}_0^{(m)})]^{1/(4p)} \mathbb{E}[\max\{\#\text{PD}(\mathcal{X}_0), \#\text{PD}(\mathcal{X}_0^{(m)})\}^{4q/r}]^{1/(4q)} \\
&\leq 2^{1+1/(4q)} \mathbb{E}[d_H^{4p}(\mathcal{X}_0, \mathcal{X}_0^{(m)})]^{1/(4p)} \mathbb{E}[\#\text{PD}(\mathcal{X}_0)^{4q/r}]^{1/(4q)}.
\end{aligned}$$

Now, if we take $p = r/4$ (recall that $r > 4$ is characterized in Assumption 1.1) and accordingly $q = r/(r-4)$, then

$$\sum_{m \geq 1} \mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \text{PD}(\mathcal{X}_0^{(m)}))]^{1/4}$$

$$\leq 2^{5/4-1/r} \mathbb{E}[\#\text{PD}(\mathcal{X}_0)^{4/(r-4)}]^{1/4-1/r} \sum_{m \geq 1} \mathbb{E}[d_H^r(\mathcal{X}, \mathcal{X}_0^{(m)})]^{1/r}$$

and the RHS is finite by assumption. This shows that the one-dimensional processes $X_i = W_r^2(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}})$ and $Y_i = W_r^2(\text{PD}(\mathcal{Y}_i), \mu_{\mathcal{Y}})$ enjoy the L^2 - m -approximation property.

Furthermore by assumption $\mathbb{E}[X_0] = \mathbb{E}[W_r^2(\text{PD}(\mathcal{X}_0), \mu_{\mathcal{X}})] < \infty$ as well as $\mathbb{E}[X_0^2] = \mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \mu_{\mathcal{X}})] < \infty$ and the same is true for $(Y_i)_i$. Hence the conditions of Theorem A.1 in Aue et al. [1] are satisfied for the centered processes $(X_i - \mathbb{E}[X_0])_i$, resp., $(Y_i - \mathbb{E}[Y_0])_i$ and therefore (w.r.t. the Skorokhod topology on D_1)

$$\begin{aligned} \frac{1}{\sqrt{m}} \sum_{i=1}^{\lfloor sm \rfloor} (X_i - \mathbb{E}[X_0]) &\Rightarrow \sqrt{\Gamma_{\mathcal{X}}} B_1(s), \\ \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor sn \rfloor} (X_i - \mathbb{E}[X_0]) &\Rightarrow \sqrt{\Gamma_{\mathcal{Y}}} B_2(s), \quad s \in [0, 1], \end{aligned}$$

where B_1, B_2 are independent standard Brownian motions and

$$\Gamma_{\mathcal{X}} = \sum_{i \in \mathbb{Z}} \text{Cov}(X_0, X_i), \quad \Gamma_{\mathcal{Y}} = \sum_{i \in \mathbb{Z}} \text{Cov}(Y_0, Y_i).$$

Moreover,

$$\frac{1}{m} \sum_{i=1}^m X_i \rightarrow \mathbb{E}[X_0] \quad a.s. \text{ and } \frac{1}{n} \sum_{i=1}^n Y_i \rightarrow \mathbb{E}[Y_0] \quad a.s.$$

by Corollary 3.3. Altogether, we see that $(X_i)_i$ and $(Y_i)_i$ fulfill all requirements in Theorem 3.1.

Step 3. Finally, we verify that the weak convergence carries over to the original processes. To that end, define

$$\begin{aligned} \widetilde{\text{Var}}(\mathcal{X})(s) &= \frac{1}{\lfloor ms \rfloor} \sum_{i=1}^{\lfloor ms \rfloor} X_i = \frac{1}{\lfloor ms \rfloor} \sum_{i=1}^{\lfloor ms \rfloor} W_r^2(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}}), \quad s \in [0, 1], \\ \widetilde{\text{Var}}(\mathcal{Y})(s) &= \frac{1}{\lfloor ns \rfloor} \sum_{i=1}^{\lfloor ns \rfloor} Y_i = \frac{1}{\lfloor ns \rfloor} \sum_{i=1}^{\lfloor ns \rfloor} W_r^2(\text{PD}(\mathcal{Y}_i), \mu_{\mathcal{Y}}), \quad s \in [0, 1], \end{aligned}$$

Next, set

$$\tilde{D}_{m,n}(s) = s(\widetilde{\text{Var}}(\mathcal{X})(s) - \widetilde{\text{Var}}(\mathcal{Y})(s)), \quad s \in [0, 1],$$

and $\tilde{D}_{m,n} = \tilde{D}_{m,n}(1)$ as well as

$$\tilde{V}_{m,n} = \left\{ \int_0^1 (\tilde{D}_{m,n}^2(s) - s^2 \tilde{D}_{m,n}^2 \nu(ds)) \right\}^{1/2}.$$

Then

$$\sqrt{m+n}(\widehat{V}_{m,n} - \widetilde{V}_{m,n}) \leq \sqrt{2} \sqrt{m+n} \max_{s \in [0,1]} |\widehat{D}_{m,n}^2(s) - \widetilde{D}_{m,n}^2(s)|$$

by the reverse triangle-inequality and

$$\sqrt{m+n} \left((\widehat{D}_{m,n}^2 - D^2) - (\widetilde{D}_{m,n}^2 - D^2) \right) \leq \sqrt{m+n} \max_{s \in [0,1]} |\widehat{D}_{m,n}^2(s) - \widetilde{D}_{m,n}^2(s)|.$$

Now, we have

$$\begin{aligned} & \sqrt{m} \max_{s \in [0,1]} |\widehat{D}_{m,n}^2(s) - \widetilde{D}_{m,n}^2(s)| \\ & \leq \left\{ \max_{s \in [0,1]} \sqrt{ms} \cdot |\widetilde{\text{Var}}(\mathcal{X})(s) - \widehat{\text{Var}}(\mathcal{X})(s)| + \max_{s \in [0,1]} \sqrt{ms} \cdot |\widetilde{\text{Var}}(\mathcal{Y})(s) - \widehat{\text{Var}}(\mathcal{Y})(s)| \right\} \\ & \quad \cdot \left\{ \max_{s \in [0,1]} s \cdot |\widetilde{\text{Var}}(\mathcal{X})(s) + \widehat{\text{Var}}(\mathcal{X})(s)| + \max_{s \in [0,1]} s \cdot |\widetilde{\text{Var}}(\mathcal{Y})(s) + \widehat{\text{Var}}(\mathcal{Y})(s)| \right\}. \end{aligned}$$

Consequently, the desired weak convergence for the random vector $\sqrt{m+n}(\widehat{D}_{m,n}^2 - D^2, \widetilde{V}_{m,n})$ follows if we show: For each $\delta, \varepsilon > 0$, there is a $C \in \mathbb{R}_+$ and $m_0 \in \mathbb{N}$ such that

$$\begin{aligned} \mathbb{P} \left(\max_{s \in [0,1]} s \cdot |\widetilde{\text{Var}}(\mathcal{X})(s) + \widehat{\text{Var}}(\mathcal{X})(s)| > C \right) &\leq \varepsilon, \\ \mathbb{P} \left(\max_{s \in [0,1]} s \cdot |\widetilde{\text{Var}}(\mathcal{Y})(s) + \widehat{\text{Var}}(\mathcal{Y})(s)| > C \right) &\leq \varepsilon \quad \forall m \geq m_0 \end{aligned} \quad (3.1)$$

and

$$\begin{aligned} \mathbb{P} \left(\max_{s \in [0,1]} \sqrt{ms} \cdot |\widetilde{\text{Var}}(\mathcal{X})(s) - \widehat{\text{Var}}(\mathcal{X})(s)| > \delta \right) &\leq \varepsilon, \\ \mathbb{P} \left(\max_{s \in [0,1]} \sqrt{ms} \cdot |\widetilde{\text{Var}}(\mathcal{Y})(s) - \widehat{\text{Var}}(\mathcal{Y})(s)| > \delta \right) &\leq \varepsilon \quad \forall m \geq m_0. \end{aligned} \quad (3.2)$$

Clearly, it is sufficient to focus on the case for $\mathcal{X}$. We treat (3.1) first and (3.2) afterwards. To this end, we bound above the summands of $\widetilde{\text{Var}}(\mathcal{X})(s) + \widehat{\text{Var}}(\mathcal{X})(s)$ by

$$|W_r^2(\text{PD}(\mathcal{X}_i), \widehat{\mu}_{\mathcal{X}}) + W_r^2(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}})| \leq (3W_r^2(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}}) + 2W_r^2(\widehat{\mu}_{\mathcal{X}}, \mu_{\mathcal{X}})).$$

Thus, the left-hand side of the event in (3.1) is at most

$$\begin{aligned} & \max_{s \in [0,1]} s \cdot \frac{1}{[ms]} \sum_{i=1}^{[ms]} \left(3W_r^2(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}}) + 2W_r^2(\widehat{\mu}_{\mathcal{X}}, \mu_{\mathcal{X}}) \right) \\ & \lesssim \frac{1}{m} \sum_{i=1}^m \left(3W_r^2(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}}) + 2W_r^2(\widehat{\mu}_{\mathcal{X}}, \mu_{\mathcal{X}}) \right). \end{aligned}$$